

Alexandre Marques Dias

Graphics Specialist with experience in
GPU-compute, Shaders and Performance

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Languages : French (native), English (fluent), Portuguese (fluent)

alexmdias.github.io

Programming languages & Toolset

C++: Vulkan, OpenGL, GLSL, LibiGL, Skia, WebAssembly, Emscripten and ThreeJS (with **Javascript**)

Python : OpenCV, NumPy, Pandas, Boto3, Dash, Scikit-Learn, BluePy, TensorFlow

AI-assisted development : 3 years of experience using LLMs in a professional setting for feature development, debugging and performance improvements with AI-assisted benchmarks

Experience

Pace Aerospace - C++/OpenGL Embedded Graphics Programmer (Permanent)

| September 2025 - September 2026 - Full-time

Developing a DO-178 safety-critical graphics-editing HMI engine for the military and aviation industry.

Tales of War - C# Unity Graphics programmer and consultant (Contract)

| March 2025 - June 2025 - Full-time

Graphics expert consultant for a multiplayer game at the Indie Asylum. Contracted to profile performance bottlenecks and optimize the game. Implemented a variety of optimization techniques to boost the game's performance to be run on low end hardware and on a multitude of various platforms.

Cosmographic - C++/Vulkan Graphics programmer (Graduate Internship)

| March 2024 - December 2024 - Full-time

Graphics programmer for [Space Engine](#), the most scientifically accurate space simulation. I was the Compute Shaders subject matter expert on the rendering team, developing advanced GPU-based computer graphics features for the engine using Vulkan, using a custom Vulkan RHI I helped to implement.

My Master's Thesis was written while working there. [See it here](#).

Corel Corporation - C++ Graphics R&D developer (Permanent)

| June 2022 - January 2023 - Full-time then part-time

Employed to work on the CorelDraw graphics suite. I developed typography rendering technology with the Skia graphics engine (vector rendering, language encoding, font encoding, glyphs) And was in charge of transitioning our graphics APIs from DirectX (windows) and Meta (MacOS) to Skia (Web)

Aquassist, Magnus Produits Chimiques - Python developer for chemical devices (Internship)

| May 2021 - August 2021 - Full-time

National Research Council Canada (NRC) - AI Developer for a conversational agent in Unity

| June 2020 - June 2021 - Part-time

Education

University of Montreal, - Masters of Computer Science (in computer graphics & vision) - 2022-2024

Graduated with Cum Laude distinction

University of Montreal, - Bachelor of Computer Science - 2019-2022

Graduated with Cum Laude distinction

Other Skills : Visual arts [Portfolio](#)